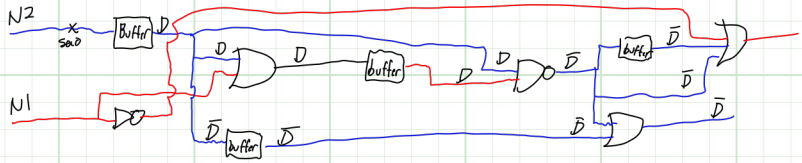
PART A

1)

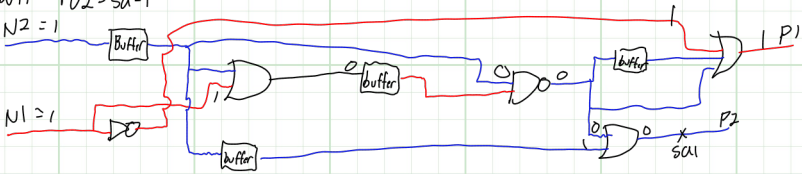


3) fault N2-sa-0



N2 : 1
N1 : X

fault PO2-sa-1



N2 : 0
N1 : 0

going through all Patterns PO2-sa1 is Undetectable

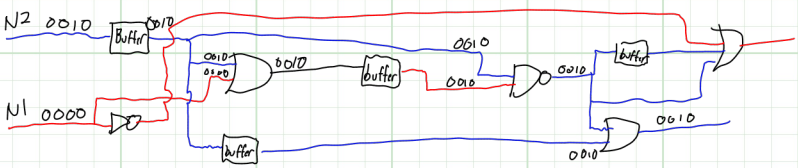
4)

```
TEST-T> report_patterns -all
  Pattern 0 (basic_scan)
  Time 0: force_all_pis =    11
  Time 1: measure_all_pos = 01
  Pattern 1 (basic_scan)
  Time 0: force_all_pis =    01
  Time 1: measure_all_pos = 11
  Pattern 2 (basic_scan)
  Time 0: force_all_pis =    10
  Time 1: measure_all_pos = 11
```

```
TEST-T> report_patterns -all
  Pattern 0 (basic_scan)
  Time 0: force_all_pis =    11
  Time 1: measure_all_pos = 01
  Pattern 1 (basic_scan)
  Time 0: force_all_pis =    01
  Time 1: measure_all_pos = 11
  Pattern 2 (basic_scan)
  Time 0: force_all_pis =    10
  Time 1: measure_all_pos = 11
```

5)

N2 sa0



P02-Sa1 is undetectable ∴ no sequence possible

6)

```
TEST-T> report_patterns -all
  Pattern 0 (full_sequential)
  Time 0: period = 100
  Time 0: force_all_pis =   011
  Time 140: measure_all_pos =XX
  Time 200: force_all_pis = 111
  Time 340: measure_all_pos =XX
  Time 400: force_all_pis = 010
  Time 540: measure_all_pos =XX
  Time 600: force_all_pis = 100
  Time 740: measure_all_pos =11
  Time 800: force_all_pis = 011
  Time 940: measure_all_pos =11
  Pattern 1 (full_sequential)
  Time 0: period = 100
  Time 0: force_all_pis =   010
  Time 140: measure_all_pos =11
  Time 200: force_all_pis = 110
  Time 340: measure_all_pos =11
  Time 400: force_all_pis = 011
  Time 540: measure_all_pos =11
  Time 600: force_all_pis = 100
  Time 740: measure_all_pos =10
```

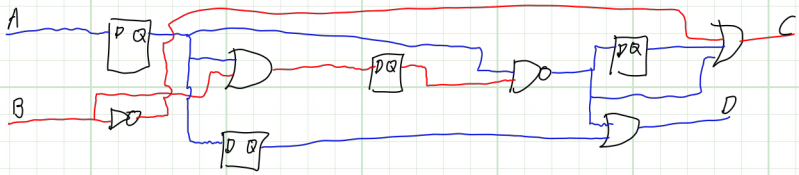
```
Time 0: period = 100
Time 0: force_all_pis =   011
Time 140: measure_all_pos =10
Time 200: force_all_pis = 100
Time 340: measure_all_pos =11
Time 400: force_all_pis = 011
Time 540: measure_all_pos =01
Time 600: force_all_pis = 101
Time 740: measure_all_pos =11
Time 800: force_all_pis = 011
Time 940: measure_all_pos =01
Pattern 3 (full_sequential)
Time 0: period = 100
Time 0: force_all_pis =   010
Time 140: measure_all_pos =01
Time 200: force_all_pis = 101
Time 340: measure_all_pos =11
Time 400: force_all_pis = 001
Time 540: measure_all_pos =11
Time 600: force_all_pis = 100
Time 740: measure_all_pos =11
```

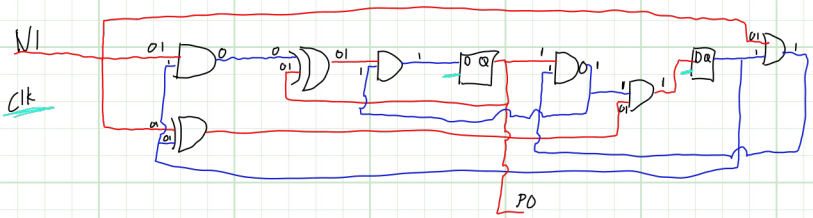
```
Pattern 4 (full_sequential)
Time 0: period = 100
Time 0: force_all_pis =   011
Time 140: measure_all_pos =11
Time 200: force_all_pis = 111
Time 340: measure_all_pos =11
Time 400: force_all_pis = 001
Time 540: measure_all_pos =11
Time 600: force_all_pis = 110
Time 740: measure_all_pos =01
```

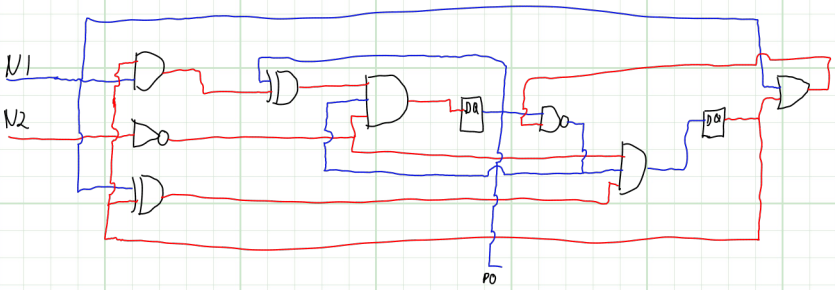
7) there are no differences for the results between the two circuits

PART B

1)







2)

Circuit 1

sa0	NC	clk
sa1	NC	clk
sa0	NC	N1
sa1	NC	N1
sa0	NC	ff1_reg/D
sa1	NC	ff1_reg/CP
sa0	NC	ff1_reg/CP
sa1	NC	ff1_reg/D
sa0	NC	ff4_reg/CP
sa1	NC	ff4_reg/D
sa0	NC	ff4_reg/D
sa1	NC	ff4_reg/CP
sa0	NC	ff2_reg/CP
sa0	NC	U9/A2
sa0	NC	U9/A1
sa1	NC	U9/Z
sa0	NC	U9/Z
sa1	NC	ff2_reg/CP
sa0	NC	ff3_reg/CP
sa1	NC	ff3_reg/D
sa0	NC	ff3_reg/D
sa1	NC	ff3_reg/CP
sa0	NC	ff1_reg/Q
sa1	NC	ff1_reg/Q
sa0	NC	U8/ZN
sa1	NC	U8/A2
sa1	NC	ff2_reg/Q
sa1	NC	U8/ZN
sa0	NC	PO2
sa0	NC	U10/A2
sa0	NC	ff4_reg/Q
sa1	NC	PO2
sa0	NC	U11/ZN
sa0	NC	U12/A3
sa0	NC	ff3_reg/Q
sa0	NC	PO1
sa1	NC	PO1

Circuit 2

sa0	NC	clk
sa1	NC	clk
sa0	NC	CNT
sa1	NC	CNT
sa0	NC	ff1_reg/CP
sa1	NC	U1/A2
sa0	NC	U1/A2
sa1	NC	ff1_reg/CP
sa0	NC	ff2_reg/CP
sa1	NC	U5/A2
sa1	NC	ff2_reg/CP
sa0	NC	U2/A2
sa0	NC	PO
sa1	NC	PO
sa0	AN	U3/Z
sa1	AN	ff1_reg/Q
sa0	AN	ff1_reg/Q
sa1	AN	U3/A2
sa1	AN	U3/Z
sa1	AN	U1/A1
sa0	AN	U1/A1
sa1	AN	U1/Z
sa1	AN	U4/ZN
sa1	AN	U4/A2
sa0	AN	U4/ZN
sa0	AN	U2/A1
sa1	AN	U2/Z
sa0	AN	U5/Z
sa1	AN	ff2_reg/Q
sa0	AN	ff2_reg/Q
sa1	AN	U7/A2
sa1	AN	U7/Z
sa1	AN	U6/A1
sa0	AN	U6/A1
sa1	AN	U6/Z
sa0	AN	U7/Z
sa1	AN	U5/A1
sa1	AN	U5/Z

Circuit 3

sa0	NC	clk
sa1	NC	clk
sa0	NC	CNT
sa1	NC	CNT
sa1	NC	U1/ZN
sa0	NC	U1/ZN
sa0	NC	ff1_reg/CP
sa1	NC	ff1_reg/Q
sa0	NC	ff1_reg/Q
sa1	NC	U4/A3
sa1	NC	U4/A2
sa1	NC	U4/Z
sa1	NC	U2/A2
sa0	NC	U2/A2
sa1	NC	U2/A1
sa0	NC	U2/A1
sa1	NC	U2/Z
sa0	NC	U4/Z
sa1	NC	ff1_reg/CP
sa0	NC	ff2_reg/CP
sa1	NC	ff2_reg/Q
sa0	NC	ff2_reg/Q
sa1	NC	U8/A3
sa1	NC	U8/A2
sa1	NC	U8/Z
sa1	NC	U7/A1
sa0	NC	U7/A1
sa1	NC	U7/Z
sa0	NC	U8/Z
sa1	NC	U6/A2
sa1	NC	U6/A1
sa1	NC	U6/Z
sa0	NC	U6/Z
sa1	NC	ff2_reg/CP
sa1	NC	U5/ZN
sa1	NC	U5/A2
sa0	NC	U5/ZN
sa0	NC	U3/A2
sa0	NC	U3/A1
sa1	NC	U3/Z
sa0	NC	PO
sa1	NC	PO

3) look at stil files

4) amount of flipflops and gates
w/ circuit 3 adding a clr signal

5) basic covers combinational logic well
fast covers more faults
full covers most faults by doing multi cylces

Circuit 1 harder to test sequential fault
Circuit 2 xor is hard to propagate faults
Circuit 3 clr signal is hard to dechipher faults

6) Circuit 1 has differences because of ff's
Circuit 2 has differences because of XOR's
Circuit 3 has differences because of CLR signal